

一. 选择题

1. D 2. B 3. A

二.

(1) 利用球坐标求元

$$\iiint_V z \, dv = \int_0^{2\pi} d\varphi \int_0^{\frac{\pi}{2}} d\varphi \int_0^{2\cos\varphi} r \cos\varphi \, r^2 \sin\varphi \, dr$$

$$\begin{cases} x^2 + y^2 + z^2 \leq 2za \\ x^2 + y^2 \leq z^2 \end{cases} \Rightarrow \begin{cases} r^2 \leq 2r \cos\varphi \\ r^2 \sin^2\varphi \leq r^2 \cos^2\varphi \end{cases}$$

$$\Rightarrow r \in [0, 2\cos\varphi], \varphi \in [0, \frac{\pi}{4}]$$

$$I = \frac{2\pi}{2} \int_0^{\frac{\pi}{4}} \cos 2\varphi \, d\varphi \int_0^{2\cos\varphi} r^3 \, dr$$

$$= \pi \int_0^{\frac{\pi}{4}} \cos 2\varphi \, d\varphi \left[\frac{1}{4} (2\cos\varphi)^4 \right] \quad 2\cos^2\varphi - 1 = \cos 2\varphi$$

$$= \pi \int_0^{\frac{\pi}{4}} a^4 \cos\varphi \cos 2\varphi \, d\varphi$$

$$= \pi a^4 \int_0^{\frac{\pi}{4}} \cos\varphi \left(\frac{\cos\varphi + 1}{2} \right)^2 d\varphi$$

$$= \pi a^4 \times \frac{7}{20} = \frac{7}{10} \pi a^4$$

$$(2) \iiint_V xyz \, dv = \frac{1}{48}$$

$$x^2 + y^2 + z^2 = 1 \Rightarrow r \in (0, 1) \quad \varphi \in (0, \frac{\pi}{2}), \theta \in (0, 2\pi)$$

$$\iiint_V xyz \, dv = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 r^2 \sin\varphi \, r^3 \cos\varphi \sin\theta \sin^2\varphi \cos\varphi \, dr$$

$$= \int_0^{2\pi} \sin\theta \cos\theta \, d\theta \int_0^{\frac{\pi}{2}} \sin^3\varphi \cos\varphi \, d\varphi \int_0^1 r^5 \, dr$$

$$= \frac{1}{6} \times \frac{1}{2} \times \int_0^{\frac{\pi}{2}} \sin^3\varphi \, d\varphi$$

$$= \frac{1}{6} \times \frac{1}{2} \left[-\sin^2\varphi \right]_0^{\frac{\pi}{2}} \times \frac{1}{2} = \frac{1}{48}$$

$$(1) \iiint_V (x^2 + y^2 + z^2) dv$$

$$\frac{x^2+y^2}{3} \leq z \leq 3$$

$$(x, y, z) \rightarrow (r, \theta, z)$$

$$\frac{r^2}{3} \leq z \leq 3$$

$$I = \int_0^{2\pi} d\theta \int_0^3 r dr \int_{\frac{r^2}{3}}^3 (r^2 + z^2) dz$$

$$= \int_0^{2\pi} d\theta \int_0^3 r dr \left(r^2 \frac{1}{3} z^3 \right) \Big|_{\frac{r^2}{3}}^3$$

$$= \int_0^{2\pi} d\theta \int_0^3 r \left(r^3 + \frac{1}{3} r^2 (27 - \frac{r^6}{27}) \right) dr$$

$$= 2\pi \left(\frac{1}{4} r^4 + \frac{9}{2} r^2 - \frac{r^8}{8 \times 27} \right) \Big|_0^3$$

$$= 2\pi \left(\frac{81}{4} + \frac{81}{2} - \frac{3^8}{8 \times 27} \right)$$

$$= 2\pi \left(\frac{81}{4} + \frac{81}{2} - \frac{81}{8} \right)$$

$$= 2\pi \left(\frac{405}{8} \right) = \frac{405}{4} \pi$$

$$(3) (x, y, z) \rightarrow (r, \theta, z)$$

$$r \in [0, 1]$$

$$r \cos \varphi \geq r \sin \varphi \Rightarrow \varphi \in [0, \frac{\pi}{4}]$$

$$\theta \in [0, 2\pi]$$

$$I = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^1 e^r r^2 \sin \varphi dr$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} \sin \varphi d\varphi \int_0^1 e^r r^2 dr$$

$$\int_0^1 e^r r^2 dr = \int_0^1 r^2 d(e^r) = r^2 e^r \Big|_0^1 - \int_0^1 e^r dr$$

$$= e - \int_0^1 r e^r dr$$

$$= e - \int_0^1 r d(e^r) = e - r e^r \Big|_0^1 + \int_0^1 e^r dr$$

$$= e - 2 + 2 e^r \Big|_0^1$$

$$\therefore I = (e-2) 2\pi \left(\cos \varphi \Big|_0^{\frac{\pi}{4}} \right)$$

$$= (e-2) 2\pi \left(1 - \frac{\sqrt{2}}{2} \right) = \pi (2-\sqrt{2}) (e-2)$$

$$(2) (x, y, z) \rightarrow (r, \theta, z)$$

$$\frac{r}{\sqrt{5}} \leq z \leq \sqrt{5}, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 3$$

$$I = \int_0^{2\pi} d\theta \int_0^3 r dr \int_{\frac{r}{\sqrt{5}}}^{\sqrt{5}} r^2 z^2 dz$$

$$= \int_0^{2\pi} d\theta \int_0^3 r dr \left(\frac{1}{3} r^2 z^3 \right) \Big|_{\frac{r}{\sqrt{5}}}^{\sqrt{5}}$$

$$= \int_0^{2\pi} d\theta \int_0^3 r dr \left(\sqrt{5} r^2 - \frac{r^5}{9\sqrt{5}} \right)$$

$$= 2\pi \int_0^3 \left(\sqrt{5} r^3 - \frac{r^6}{9\sqrt{5}} \right) dr$$

$$= 2\pi \left(\frac{\sqrt{5}}{4} r^4 - \frac{r^7}{54\sqrt{5}} \right) \Big|_0^3$$

$$= 2\pi \left(\frac{81\sqrt{5}}{4} - \frac{3^7}{54\sqrt{5}} \right)$$

$$= 2\pi \left(\frac{81\sqrt{5}}{4} - \frac{27}{2\sqrt{5}} \right)$$

$$= 2\pi \frac{27}{4} \sqrt{5} = \frac{27}{2} \sqrt{5} \pi$$

$$(4) x, y, z \text{ 用球坐标代换}$$

$$x^2 + y^2 + z^2 = r^2 = r \cos \varphi \Rightarrow \cos \varphi = r$$

$$I = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin \varphi d\varphi \int_0^{\cos \varphi} r^2 r dr$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin \varphi d\varphi \left(\frac{1}{4} \cos^4 \varphi - 0 \right)$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin \varphi \cos^4 \varphi d\varphi$$

$$= -\frac{\pi}{2} \int_0^{\frac{\pi}{2}} \cos^4 \varphi d(\cos \varphi)$$

$$= -\frac{\pi}{2} \left(\frac{1}{5} \cos^5 \varphi \Big|_0^{\frac{\pi}{2}} \right) = -\frac{\pi}{2} \times \frac{1}{5} \times (0 - 1)$$

$$= \frac{\pi}{10}$$

IV. $(x-1)^2 + (y-2)^2 + (z-3)^2 \leq 1$

$$\begin{cases} x = 1 + r \sin \varphi \cos \theta \\ y = 2 + r \sin \varphi \sin \theta \\ z = 3 + r \cos \varphi \end{cases}$$

$$dx dy dz = r^2 \sin \varphi dr d\varphi d\theta$$

$$V = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin \varphi d\varphi \int_0^1 (r \sin \varphi \cos \theta)^{\frac{2}{3}} dr$$

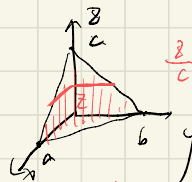
$$= \int_0^{2\pi} (\cos \theta)^{\frac{2}{3}} d\theta \int_0^{\frac{\pi}{2}} (\sin \varphi)^{\frac{5}{3}} d\varphi \int_0^1 r^{\frac{8}{3}} dr$$

$$= \frac{3}{11} \times \int_0^{2\pi} (\cos^{\frac{2}{3}} \theta) d\theta \int_0^{\frac{\pi}{2}} (\sin \varphi)^{\frac{5}{3}} d\varphi$$

$$= -\frac{3}{11} \times \int_0^{2\pi} (\cos^{\frac{2}{3}} \theta) d\theta \int_0^{\frac{\pi}{2}} (1 - \cos^2 \varphi)^{\frac{1}{3}} d(\cos \varphi)$$

$$= -\frac{3}{11} \times \frac{-12}{5} \pi = \frac{36}{55} \pi$$

V. (1) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$



or Δabc base

$$S_{abc} = \frac{3V_{abc}}{h}$$

$$h = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$V_{abc} = \iiint dx dy dz$$

$$= \int_0^c dz \iint dy dx$$

$$= \int_0^c \frac{z^2}{c^2} \frac{db}{2} dz$$

$$= \frac{ab}{2c^2} \int_0^c z^2 dz = \frac{1}{6} abc$$

$$S_{abc} = \frac{\frac{1}{6} abc}{h} = \frac{abc \sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}{2}$$

VI. (1) $z = 6 - x^2 - y^2$, $z = \sqrt{x^2 + y^2}$

用柱坐标

$$z = 6 - r^2, z = r$$

$$\Rightarrow r \in [0, 2]$$

$$\theta \in [0, 2\pi]$$

$$z \in [r, 6 - r^2]$$

$$V = \int_0^{2\pi} d\theta \int_0^2 r dr \int_r^{6-r^2} dz$$

$$= 2\pi \int_0^2 (6r - r^2 - r^3) dr$$

$$= \frac{32}{3} \pi$$

(2) $z = \sqrt{5 - x^2 - y^2}$, $x^2 + y^2 = 4z$

用柱坐标

$$z = \sqrt{5 - r^2}, z = \frac{r^2}{4}$$

$$\sqrt{1 - \frac{r^2}{5}}$$

$$\Rightarrow \theta \in [0, 2\pi], r \in [0, 4], z \in [\frac{r^2}{4}, \sqrt{5 - r^2}]$$

$$V = \int_0^{2\pi} d\theta \int_0^4 r dr \int_{\frac{r^2}{4}}^{\sqrt{5 - r^2}} dz = 2\pi \int_0^4 (\frac{r^3}{4} - r\sqrt{5 - r^2}) dr$$

$$= 2\pi \int_0^4 \frac{r^3}{8} dr^2 - 2\pi \int_0^4 \sqrt{5 - r^2} dr^2 = \frac{2}{3} \pi (5\sqrt{5} - 4)$$

(2) $z = \sqrt{x^2 + y^2}$, $z^2 = 2x$

$$\theta \in [0, \frac{\pi}{2}], r \in [0, 2\cos \theta]$$

$$x^2 + y^2 = 2x \Rightarrow (x-1)^2 + y^2 = 1$$

$$r^2 = 2r \cos \theta$$

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\sqrt{1 + z_x^2 + z_y^2} = \sqrt{2}$$

$$V = \iint_D \sqrt{2} dx dy = 2 \int_0^{\frac{\pi}{2}} d\theta \int_0^{2\cos \theta} r dr$$

$$= 2 \int_0^{\frac{\pi}{2}} d\theta \left. \frac{r^2}{2} \right|_0^{2\cos \theta}$$

$$= 2 \int_0^{\frac{\pi}{2}} 2\cos^2 \theta d\theta$$

$$= 2 \int_0^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta$$

$$= 2\theta \Big|_0^{\frac{\pi}{2}} + \sin 2\theta \Big|_0^{\frac{\pi}{2}} = \pi + 0 = \pi$$

$$= \sqrt{2} \pi + 0 = \sqrt{2} \pi$$